Diagnoses, Admission Type, and Readmission

Diabetes 130-US Hospitals (1999-2008)

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CAI5735: Applied Data Science in Health

Research Question

How is the number of diagnoses related to hospital readmission, and does that relationship differ by admission type (Emergency, Urgent, Elective)?

Columns: number_diagnoses, readmitted, admission_type

Dataset

3,000 hospital stays of patients with diabetes (course sample, no missing values)

Source: UCI Machine Learning Repository; Strack et al. (2014), doi:10.24432/C5230J

number_diagnoses: diagnoses recorded for the stay

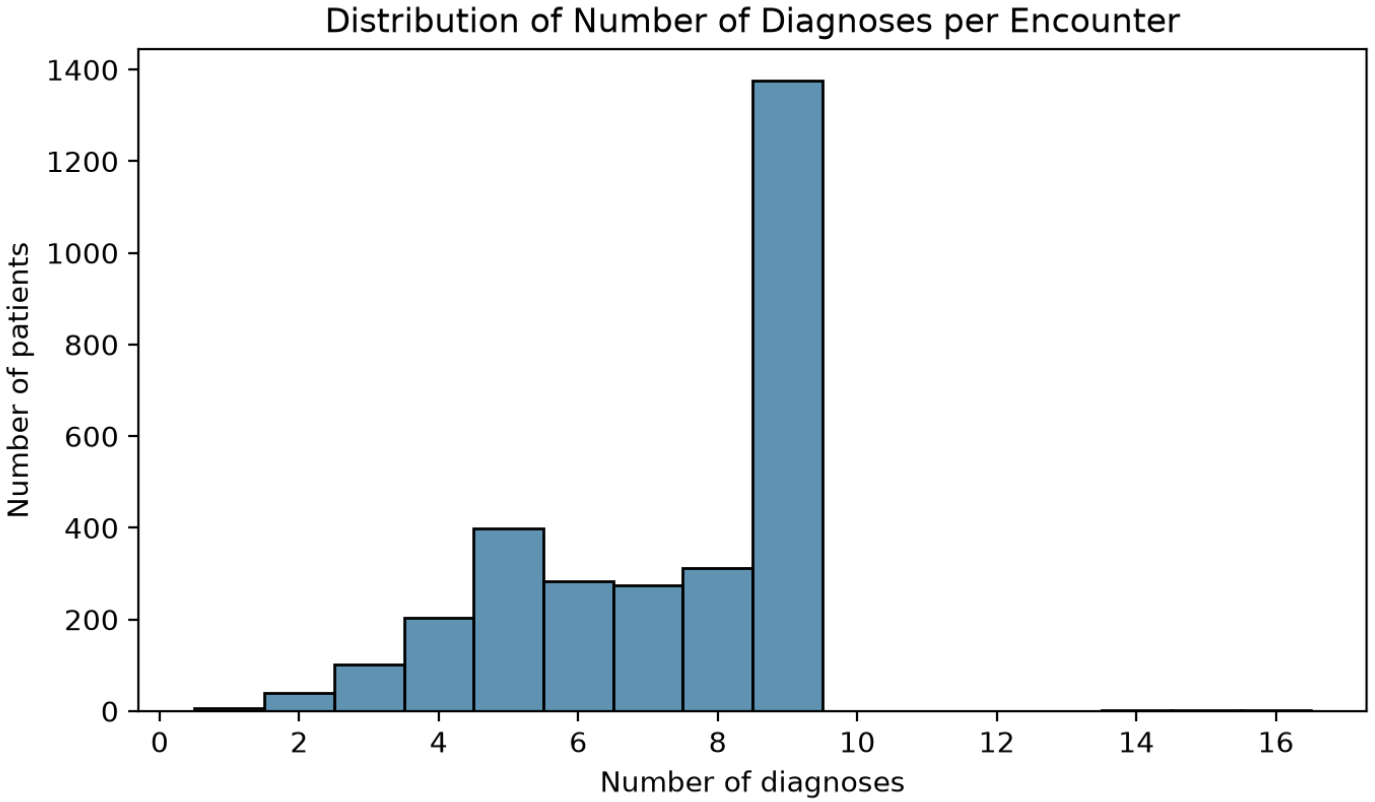
readmitted: not readmitted (60%), after 30 days (31%), within 30 days (9%)

admission_type: Emergency (57%), Elective (22%), Urgent (21%)

Analysis Process

1. Kept the three relevant columns
2. Computed columns: readmitted flag (0/1) and diagnosis group (1-4, 5-6, 7-8, 9+)
3. Filtered to Emergency admissions as a check within one type
4. Grouped by admission type and diagnosis group; sorted by readmission rate
5. Visualized distribution, outcome differences, and group trends

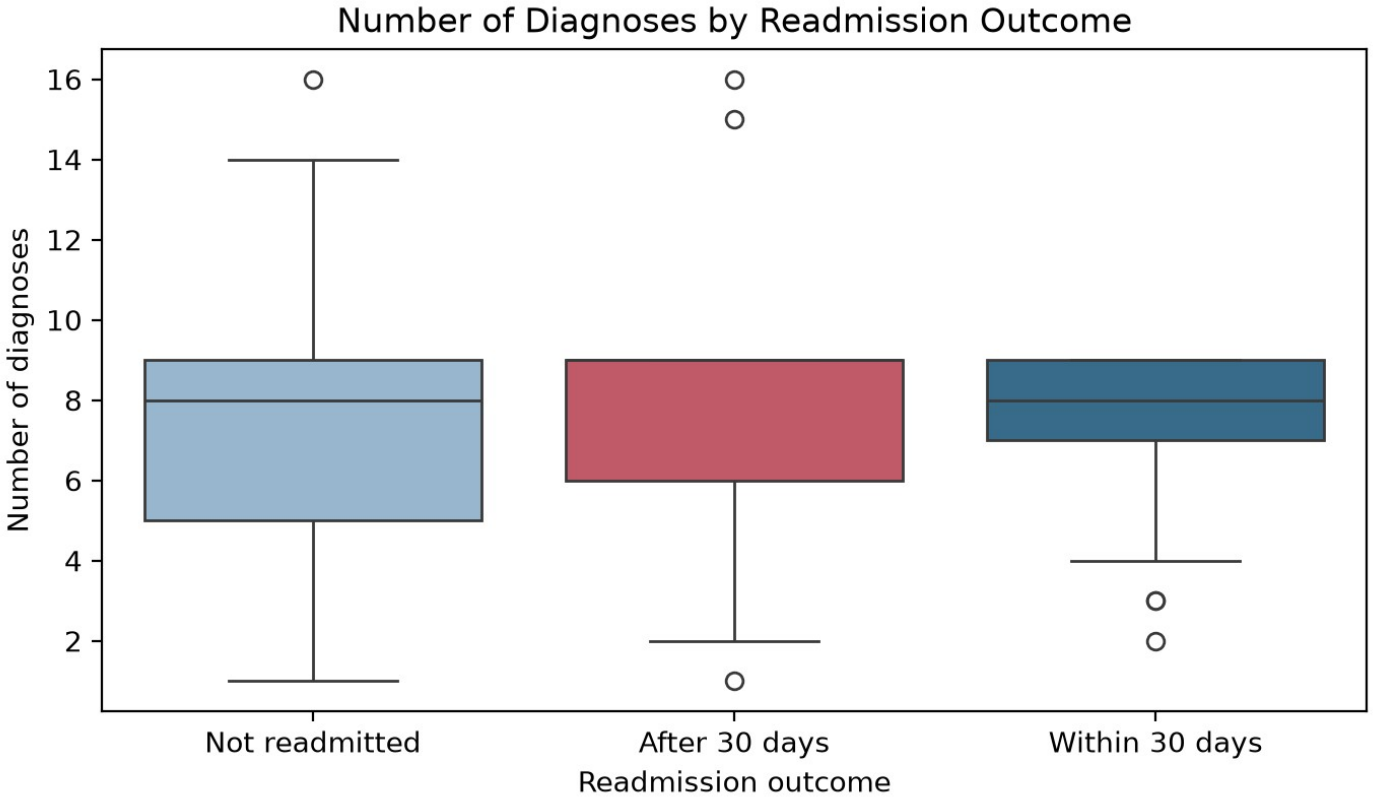
Most stays have many diagnoses



Most common value is 9 (1,376 of 3,000 stays)

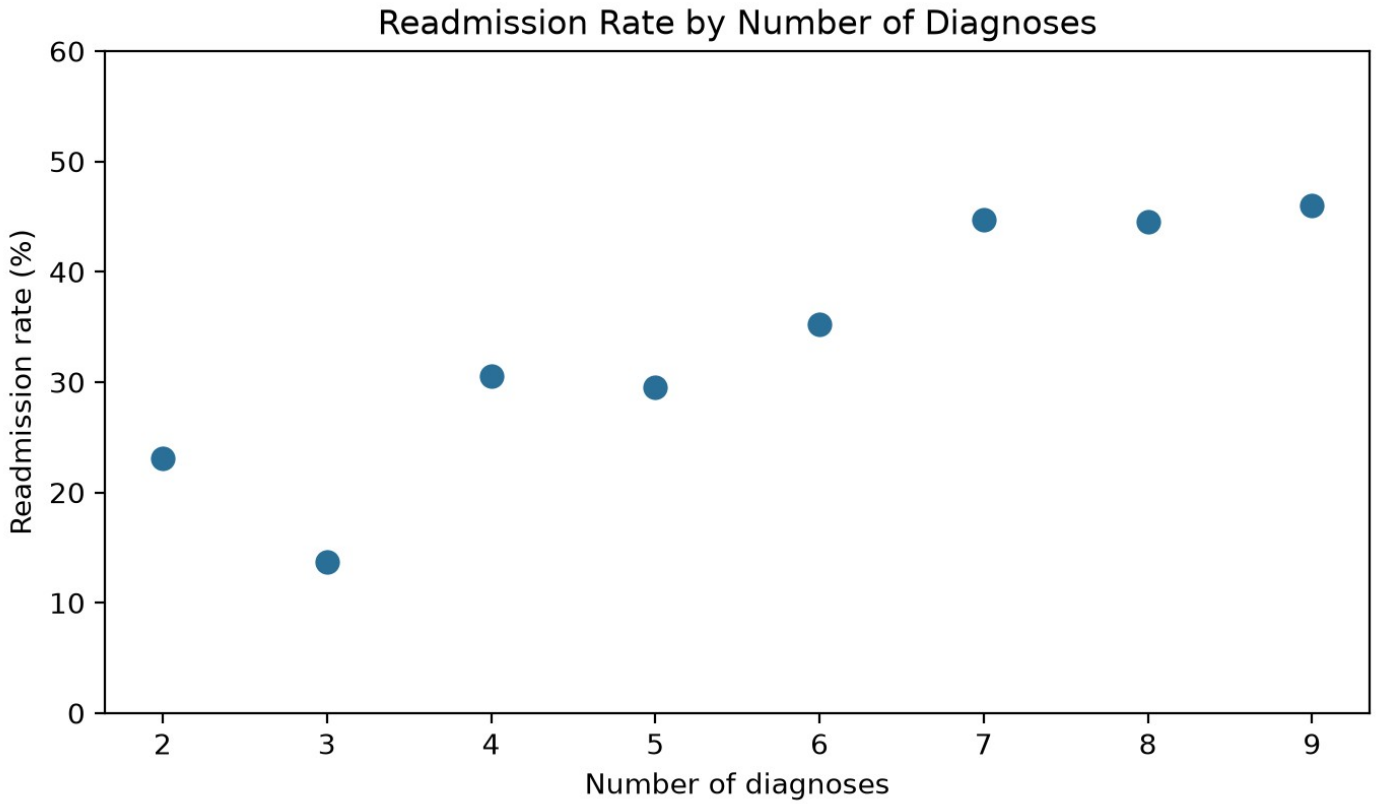
Likely a recording cap, so heavy cases may be undercounted

Readmitted patients have more diagnoses



Mean diagnoses: 7.0 not readmitted vs 7.7 readmitted (either timing)

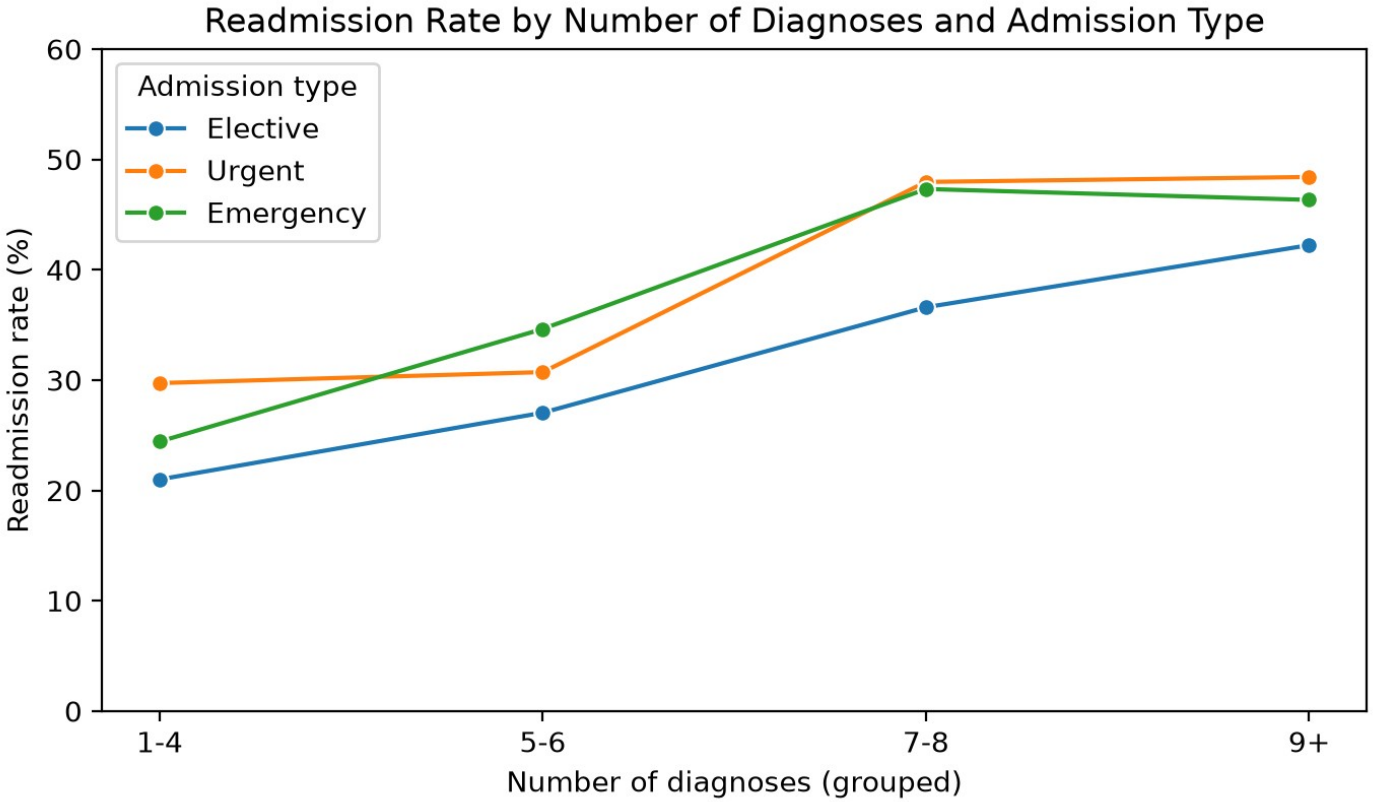
Readmission rises with diagnoses



Rate climbs from 14-23% at 2-3 diagnoses to 46% at 9

Overall: 40.0% readmitted

Same pattern in every admission type



1-4 dx: 24.6% overall; 9+ dx: 46.0%

Elective is lowest in every group (34.1% overall vs 41.4% Urgent, 41.8% Emergency)

Emergency-only check shows the same rise

Key Takeaways

More diagnoses means higher readmission, leveling off after about 7

The pattern holds across Elective, Urgent, and Emergency admissions; Elective is consistently lower

Limits: association not causation; many stays have exactly 9 diagnoses; small subgroups (Urgent 1-4 has 74 patients)

Questions?

github.com/apriljwalter-lgtm/Group_8_Midterm_project